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RESEARCH ARTICLE

Characterizations of numerical semigroup complements via Apéry sets

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Abstract In this paper, we generalize the work of Tuenter to give an identity which completely characterizes the complement of a numerical semigroup in terms of its Apéry sets. Using this result, we compute the mth power Sylvester and alternating Sylvester sums for free numerical semigroups. Explicit formulas are given for small m.

Keywords Numerical semigroups · Sylvester sums · Frobenius problem · Free numerical semigroups · Smooth sequences · Non-representable numbers · Apéry sets

1 Introduction, preliminaries

Let $\mathbb{N}_0$ denote the set of non-negative integers. A *numerical semigroup* S is a subset of $\mathbb{N}_0$ that is closed under addition, contains 0, and has finite complement in $\mathbb{N}_0$. For any nonempty $G \subset \mathbb{N}_0$, let $\langle G \rangle = \{\sum_{i=1}^{k} n_i g_i : k \in \mathbb{N}_0, n_i \in \mathbb{N}_0, g_i \in G\}$, the set of all non-negative linear combinations of elements of G. It is known that $\langle G \rangle$ is a numerical semigroup if and only if $\gcd(G) = 1$. Furthermore, given any numerical semigroup S, there is some finite generating set G such that $S = \langle G \rangle$. These results

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are well known. For a comprehensive introduction to numerical semigroups, see [8, Chapter 1].

For S a numerical semigroup with $S = \langle G \rangle$, the complement of S in $\mathbb{N}_0$ is the set of positive integers that are *not representable* in terms of G, and we denote this finite set $NR = \mathbb{N}_0 \backslash S$. The *genus* of S is $g(S) = \#NR$, and the *Frobenius number* of S is $F(S) = \max(\mathbb{Z} \backslash S)$, which is $\max(NR)$ when $NR \neq \emptyset$.

The problem of computing the genus and Frobenius number of a numerical semigroup dates back to the late nineteenth century. In [10], Sylvester solved the problem when S is generated by two integers. In the twentieth century, the idea of the Apéry set of an element in a numerical semigroup was introduced in [1]. If one knows the Apéry set of an element, then one can then compute the Frobenius number and genus of the numerical semigroup. The formula for the Frobenius number is attributed to Brauer and Shockley [2], and the genus result is attributed to Selmer [9]. In general, the Apéry set of an element in a numerical semigroup is difficult to compute.

In [11], Tuenter remarks that one can recover a set if one knows the images of the set under sufficiently many mappings. In particular, he notes that the Sylvester sums of NR are sufficient to completely determine NR, and he gives an identity ([11, Theorem 2.1]) that allows for a simple and efficient computation for these sums in the case that $NR = \mathbb{N}_0 \backslash \langle a, b \rangle$ (that is, NR is the complement of a semigroup generated by two elements). In this paper, we give a reformulation of Tuenter's identity for general semigroups in terms of Apéry sets (Theorem 2.3). Apéry sets (at least for certain values in S) are well understood when S is free, thus our identity gives a complete characterization of NR for free numerical semigroups.

This paper is organized as follows. In Sect. 2, we define Apéry sets and use them to prove Theorem 2.3, the identity which characterizes NR. We then use the identity to produce a well-known formula for the Hilbert series of a numerical semigroup in terms of the Apéry set of any element in the semigroup. In Sect. 3, we describe free numerical semigroups and their Apéry sets. Such numerical semigroups include those generated by compound sequences, geometric sequences, supersymmetric sequences, and two-element sets. Finally, in Sect. 4, we give an explicit version of the identity for free numerical semigroups, and we use it to compute the mth power and alternating mth power Sylvester sums of these semigroups.

2 An identity for Apéry sets

Definition 2.1 Let S be a numerical semigroup. For any nonzero $t \in S$, the *Apéry set* of t in S is defined as

$$\mathrm{Ap}(S; t) = \{s \in S : s - t \notin S\}.$$

Equivalently, if we let $w_i = \min\{s \in S : s \equiv i \pmod{t}\}$, then $\mathrm{Ap}(S; t) = \{w_i : 0 \leq i < t\}$. In other words, $\mathrm{Ap}(S; t)$ consists of the minimal element of S in each congruence class modulo t.

Apéry sets are incredibly helpful in understanding properties of a numerical semigroup and its complement. For instance, given a numerical semigroup S and some

$t \in S$, one has the following formulas for the genus [9, p. 3] and Frobenius number [2, Lemma 3] of S in terms of $\mathrm{Ap}(S;t)$:

$$g(S) = -\frac{t-1}{2} + \frac{1}{t} \sum_{n \in \mathrm{Ap}(S;t)} n \tag{1}$$

and

$$F(S) = \max(\mathrm{Ap}(S;t)) - t. \tag{2}$$

With Apéry sets, we have an identity which can be seen as a generalization of [4, Theorem 3.3] (Corollary 3.14 in this paper), which itself is a generalization of [11, Theorem 2.1] (Corollary 2.6 in this paper). We first need a lemma.

Lemma 2.2 *Let S be a numerical semigroup with complement $NR = \mathbb{N}_0 \backslash S$. For any nonzero $t \in S$, let $(NR + t) := \{n + t : n \in NR\}$. For any $a, b \in \mathbb{Z}$, let $I_{a,b} := \{n \in \mathbb{Z} : a \leq n < b\}$. Finally, let $\mathrm{Ap}_t := \{s \in \mathrm{Ap}(S;t) : s < t\} = \mathrm{Ap}(S;t) \cap I_{0,t}$. Then*

$$(NR + t) \backslash NR = \mathrm{Ap}(S;t) \backslash \mathrm{Ap}_t \tag{3}$$

and

$$NR \backslash (NR + t) = I_{0,t} \backslash \mathrm{Ap}_t\,. \tag{4}$$

Proof To prove these equalities, we will show the sets are subsets of each other.

For Eq. (3), to show inclusion we suppose $s \in (NR + t) \backslash NR$. Note that $t > 0$. Since $s \in (NR + t)$, $s = n + t$ for some gap $n \in NR$. The gaps of S are positive integers, so $s > t$. Since $s \notin NR$ and $s > 0$, $s \in S$. Then, with $s \in S$ and $s - t \in NR$, we have $s \in \mathrm{Ap}(S;t)$. Finally, since $s > t$, $s \notin \mathrm{Ap}_t$. Thus $s \in \mathrm{Ap}(S;t) \backslash \mathrm{Ap}_t$, so $(NR + t) \backslash NR \subseteq \mathrm{Ap}(S;t) \backslash \mathrm{Ap}_t$.

For the reverse inclusion, we suppose $s \in \mathrm{Ap}(S;t) \backslash \mathrm{Ap}_t$. Since $s \in \mathrm{Ap}(S;t)$, we have $s \in S$ and $s - t \notin S$. Since $s \in \mathrm{Ap}(S;t)$ and $s \notin \mathrm{Ap}_t$, $s > t$, so $s - t > 0$. Thus $s - t \in NR$, so $s \in (NR + t)$. Finally, since $s \in S$, $s \notin NR$. Therefore $s \in (NR + t) \backslash NR$, so $\mathrm{Ap}(S;t) \backslash \mathrm{Ap}_t \subseteq (NR + t) \backslash NR$. Combined with the previous paragraph, we have $(NR + t) \backslash NR = \mathrm{Ap}(S;t) \backslash \mathrm{Ap}_t$.

For Eq. (4), to show inclusion we suppose $n \in NR \backslash (NR + t)$. Let $m = n - t$. Since $n \in NR$ and $t \in S$, we must have $m \notin S$. Since $n \notin (NR + t)$, $m \notin NR$, so $m < 0$. Thus $1 \leq n < t$. Furthermore, $n \in NR$ so $n \notin \mathrm{Ap}_t$. Thus $n \in I_{0,t} \backslash \mathrm{Ap}_t$, so $NR \backslash (NR + t) \subseteq I_{0,t} \backslash \mathrm{Ap}_t$.

For the reverse inclusion, we suppose $n \in I_{0,t} \backslash \mathrm{Ap}_t$. Since $0 \leq n < t$, if $n \in S$ then n is the minimal element of S in its congruence class modulo t, so $n \in \mathrm{Ap}(S;t)$ and therefore $n \in \mathrm{Ap}_t$. However, since $n \notin \mathrm{Ap}_t$, this is impossible so we must have $n \notin S$, so $n \in NR$. And since $n - t < 0$, we have $n \notin (NR + t)$. Therefore $n \in NR \backslash (NR + t)$, so $I_{0,t} \backslash \mathrm{Ap}_t \subseteq NR \backslash (NR + t)$. Combined with the previous paragraph, we have $NR \backslash (NR + t) = I_{0,t} \backslash \mathrm{Ap}_t$. □

We now have our main result.

Theorem 2.3 *Let S be a numerical semigroup with complement $NR = \mathbb{N}_0 \backslash S$. For any nonzero $t \in S$ and any function f defined on $\mathbb{N}_0$,*

$$\sum_{n \in NR} [f(n+t) - f(n)] = \sum_{n \in \mathrm{Ap}(S;t)} f(n) - \sum_{n=0}^{t-1} f(n). \tag{5}$$

Proof By Lemma 2.2, $(NR + t) \backslash NR = \mathrm{Ap}(S; t) \backslash \mathrm{Ap}_t$, so for any function f defined on these sets, we have

$$\sum_{n \in (NR+t) \backslash NR} f(n) = \sum_{n \in \mathrm{Ap}(S;t) \backslash \mathrm{Ap}_t} f(n). \tag{6}$$

Since Ap_t and $\mathrm{Ap}(S; t)$ are finite sets and $\mathrm{Ap}_t \subseteq \mathrm{Ap}(S; t)$, we can rewrite Eq. (6) as

$$\sum_{n \in (NR+t) \backslash NR} f(n) = \sum_{n \in \mathrm{Ap}(S;t)} f(n) - \sum_{n \in \mathrm{Ap}_t} f(n). \tag{7}$$

Similarly, by Lemma 2.2, $NR \backslash (NR + t) = I_{0,t} \backslash \mathrm{Ap}_t$, so for any function f defined on these sets, we have

$$\sum_{n \in NR \backslash (NR+t)} f(n) = \sum_{n \in I_{0,t} \backslash \mathrm{Ap}_t} f(n). \tag{8}$$

Since Ap_t and $I_{0,t}$ are finite and $\mathrm{Ap}_t \subseteq I_{0,t}$, we can rewrite Eq. (8) as

$$\sum_{n \in NR \backslash (NR+t)} f(n) = \sum_{n \in I_{0,t}} f(n) - \sum_{n \in \mathrm{Ap}_t} f(n). \tag{9}$$

Then, since

$$\sum_{n \in (NR+t)} f(n) - \sum_{n \in NR} f(n) = \sum_{n \in (NR+t) \backslash NR} f(n) - \sum_{n \in NR \backslash (NR+t)} f(n), \tag{10}$$

we simplify the right side of Eq. (10) using Eqs. (7) and (9) to conclude

$$\sum_{n \in (NR+t)} f(n) - \sum_{n \in NR} f(n) = \sum_{n \in \mathrm{Ap}(S;t)} f(n) - \sum_{n \in I_{0,t}} f(n). \tag{11}$$

Note that any function f defined on $\mathbb{N}_0$ will necessarily be defined on the sets described above. The desired result follows immediately. □

Remark 2.4 Since $0 \in \mathrm{Ap}(S; t)$, we can exclude 0 from both summations on the right to write Eq. (5) as

$$\sum_{n \in NR} [f(n+t) - f(n)] = \sum_{\substack{n \in \mathrm{Ap}(S;t) \\ n \neq 0}} f(n) - \sum_{n=1}^{t-1} f(n) \tag{12}$$

and thereby apply this identity to functions f defined on $\mathbb{N}$.

Remark 2.5 Since $\mathrm{Ap}(S; t)$ contains one element from each congruence class modulo t, we can write Eq. (5) as

$$\sum_{n \in NR} [f(n+t) - f(n)] = \sum_{n \in \mathrm{Ap}(S;t)} [f(n) - f(\overline{n})], \tag{13}$$

where $\overline{n}$ denotes the reduction of n modulo t. In other words, given $n \in \mathbb{Z}$, the number $\overline{n}$ is the least non-negative integer congruent to n modulo t. As with Eq. (12), we can remove $n = 0$ from the summation on the right if desired.

If $S = \langle G \rangle$ for $G = \{a, b\}$ with $\gcd(a, b) = 1$, one finds that $\mathrm{Ap}(S; a) = \{nb : 0 \leq n < a\}$. (See [9].) In that case, we obtain Tuenter's result.

Corollary 2.6 [11, Theorem 2.1] *For* $G = \{a, b\}$ *with* $\gcd(a, b) = 1$, *let* $NR = \mathbb{N}_0 \backslash \langle G \rangle$. *Then for any function* f *defined on* $\mathbb{N}$,

$$\sum_{n \in NR} [f(n+a) - f(n)] = \sum_{n=1}^{a-1} [f(nb) - f(n)]. \tag{14}$$

Immediately after [11, Theorem 2.1], Tuenter mentions that Eq. (14) completely characterizes the set *NR* when $G = \{a, b\}$, which is to say that all properties of *NR* can be derived from this equation. The same argument and result holds here. Equation (5) completely characterizes the set *NR* for any set G.

2.1 Applications to the genus and Hilbert series

We conclude this section with two applications of Theorem 2.3.

2.1.1 A formula for the genus

We first use $f(n) = n$ in Eq. (5) to re-derive Selmer's genus result [Eq. (1)]. We find

$$\sum_{n \in NR} t = \sum_{n \in \mathrm{Ap}(S;t)} n - \sum_{m=0}^{t-1} m,$$

so

$$g(S) = \#NR = -\frac{t-1}{2} + \frac{1}{t} \sum_{n \in \mathrm{Ap}(S;t)} n.$$

2.1.2 A formula for the Hilbert series

The Hilbert series $H_S(x)$ of a numerical semigroup S is defined to be the formal power series

$$H_S(x) = \sum_{n \in S} x^n = \frac{1}{1-x} - \sum_{n \in NR} x^n.$$

Using $f(n) = x^n$ with Eq. (5), we obtain

$$(x^t - 1) \sum_{n \in NR} x^n = \sum_{n \in \mathrm{Ap}(S;t)} x^n - \frac{1-x^t}{1-x},$$

so

$$H_S(x) = \frac{1}{1-x^t} \sum_{n \in \mathrm{Ap}(S;t)} x^n.$$

For more on Hilbert series, including this identity, see [7]. For connections to free numerical semigroups (described in the next section), see [3, Section 5.1].

3 Explicit descriptions

Let S be a free numerical semigroup, as described in [6, Section 2.2] and [8, Chapter 8, Section 4]. The Apéry sets of free numerical semigroups are well-understood, and so in this context we obtain a further specialization of Theorem 2.3 in Corollary 3.7. We then apply our result to the special case of semigroups generated by compound sequences, as were studied in [4].

3.1 Application to free numerical semigroups

Definition 3.1 For $k \in \mathbb{N}_0$, suppose $G = (g_0, \ldots, g_k)$ is a finite sequence with $g_i \in \mathbb{N}_0$ for all i. For $0 \le i \le k$, let $G_i = (g_0, \ldots, g_i)$ and $d_i = \gcd(G_i)$. Let $c_i = d_{i-1}/d_i$ for $1 \le i \le k$. If $c_i g_i \in \langle G_{i-1} \rangle$ for $1 \le i \le k$, we say G is a *smooth sequence*. If a set can be ordered to form a smooth sequence, then we say the set is a *smooth set*. We will refer to the sequence $(c_1, \ldots, c_k)$ as the *c values of* G.

Definition 3.2 [8, p. 133] Suppose S is a numerical semigroup. If $S = \langle G \rangle$ for some set G with $\gcd(G) = 1$ and which can be ordered to form a smooth sequence, then S is a *free numerical semigroup*.

The class of smooth sequences contains the class of compound sequence, which in turn contains sequences of length two, geometric sequences, and supersymmetric sequences. (References are given in [4, Section 3]). The condition that $c_i g_i \in \langle G_{i-1} \rangle$ for all i provides us with a method (Corollary 3.4) to represent any integer uniquely in terms of elements of G, which in turn allows us to identify the elements of $\mathrm{Ap}(S; g_0)$.

We quote the following from [6] (using slightly different notation here), in which S is generated by a sequence G of positive integers. We note that these results do not require G to be minimal (a condition where no proper subset of G generates the same semigroup as G), and in fact hold even if G is a sequence of non-negative integers.

Proposition 3.3 [6, Remark, p. 11] *Let $G = (g_0, \ldots, g_k)$ be a smooth sequence with c values $(c_1, \ldots, c_k)$, let $S = \langle G \rangle$, and let $s \in S$. Then s has a unique representation*

$$s = \sum_{i=0}^{k} n_i g_i$$

with $0 \leq n_i < c_i$ for $1 \leq i \leq k$.

We extend this result for all $n \in \mathbb{Z}$.

Corollary 3.4 *Suppose $G = (g_0, \ldots, g_k)$ is smooth with c values $(c_1, \ldots, c_k)$. Let $n \in \mathbb{Z}$. Then n has a unique representation*

$$n = \sum_{i=0}^{k} n_i g_i$$

where $0 \leq n_i < c_i$ for $1 \leq i \leq k$.

Proof Let $n \in \mathbb{Z}$. If $n \in S$, by Proposition 3.3, we are done.

If $n \notin S$, then there exists some $M \in \mathbb{Z}$ such that $n + Mg_0 > F(S)$ and thus $n + Mg_0 \in S$. By Proposition 3.3, there exist integers $n_1, \ldots, n_k$ where

$$n + Mg_0 = \sum_{i=0}^{k} n_i g_i,$$

with $0 \leq n_i < c_i$ for $1 \leq i \leq k$, so n has a representation

$$n = (n_0 - M)g_0 + \sum_{i=1}^{k} n_i g_i \tag{15}$$

with $0 \leq n_i < c_i$. Thus, n has at least one representation.

To show this representation is unique, suppose we have another representation $n = \sum_{i=0}^{k} m_i g_i$ with $0 \leq m_i < c_i$ for $1 \leq i \leq k$. Then

$$n + Mg_0 = (M + m_0)g_0 + \sum_{i=1}^{k} m_i g_i.$$

Since $n+Mg_0 \in S$, it has the unique representation $n+Mg_0 = \sum_{i=0}^{k} n_i g_i$. Therefore for $n+Mg_0 = Mg_0+\sum_{i=0}^{k} m_i g_i$, we have $m_0 = n_0 - M$ and $m_i = n_i$ for $1 \le i \le k$, so this is the same representation as in Eq. (15). □

Lemma 3.5 *Suppose $S = \langle G \rangle$ with G smooth. Let $n \in \mathbb{Z}$. Given the unique representation of n, we have $n \in S$ if and only if $n_0 \ge 0$.*

Proof Clearly if $n_0 \ge 0$, then $n \in S$. Otherwise if $n_0 < 0$, then $n \notin S$ by the uniqueness of representation. □

The following proposition is a reformulation of [6, Theorem 7].

Proposition 3.6 *Suppose $S = \langle G \rangle$ for G smooth. Let $n \in \mathbb{Z}$. Given the unique representation of n, we have $n \in \mathrm{Ap}(S; g_0)$ if and only if $n_0 = 0$.*

Proof We use Lemma 3.5 for the cases $n_0 < 0$, $n_0 = 0$, and $n_0 > 0$.

If $n_0 < 0$ then $n \notin S$, so $n \notin \mathrm{Ap}(S; g_0)$. If $n_0 = 0$, then $n \in S$ and $n - g_0 \notin S$, so $n \in \mathrm{Ap}(S; g_0)$. If $n_0 > 0$, then $n \in S$ and also $n - g_0 \in S$, so $n \notin \mathrm{Ap}(S; g_0)$.

Thus $n \in \mathrm{Ap}(S; g_0)$ if and only if $n_0 = 0$. □

That is, if G is smooth then we can explicitly describe the Apéry set of g_0 in S:

$$\mathrm{Ap}(S; g_0) = \left\{ \sum_{i=1}^{k} n_i g_i : 0 \le n_i < c_i \right\}.$$

This leads to an explicit version of Theorem 2.3 for smooth sequences.

Corollary 3.7 *Suppose S is a free numerical semigroup, so $S = \langle G \rangle$ for G a smooth sequence. Then for any f defined on $\mathbb{N}_0$,*

$$\sum_{n \in NR} [f(n+g_0) - f(n)] = \sum_{n_1=0}^{c_1-1} \cdots \sum_{n_k=0}^{c_k-1} f\left(\sum_{i=1}^{k} n_i g_i\right) - \sum_{n=0}^{g_0-1} f(n). \tag{16}$$

3.2 Application to numerical semigroups generated by compound sequences

For compound sequences, we follow the notation of [4]. See also [5]. Compound sequences are special cases of smooth sequences (by Corollary 3.11), so we apply the above results to compound sequences and obtain an alternate proof of [4, Theorem 3.3].

Definition 3.8 Let $k \in \mathbb{N}_0$ and $A = (a_1, \ldots, a_k)$, $B = (b_1, \ldots, b_k) \in \mathbb{N}^k$. We say (A, B) is a *suitable pair* of k-tuples if $\gcd(a_i, b_j) = 1$ for all $i \ge j$. If (A, B) is a suitable pair, then we let $g_i = b_1 \cdots b_i a_{i+1} \cdots a_k$ for $0 \le i \le k$. The sequence $G(A, B) = (g_0, \ldots, g_k)$ is a *compound sequence*. We say a set $G = \{g_0, \ldots, g_k\} \subseteq \mathbb{N}^{k+1}$ is a *compound set* if its elements can be ordered to form a compound sequence.

Proposition 3.9 [5, Proposition 2, Property 5] *If (A, B) is a suitable pair, then* $\gcd(G(A, B)) = 1$*, so $\langle G(A, B)\rangle$ is a numerical semigroup.*

Proposition 3.10 *Let $G = (g_0, \ldots, g_k) = G(A, B)$ be a compound sequence for a suitable pair (A, B) of k-tuples with $A = (a_1, \ldots, a_k)$, $B = (b_1, \ldots, b_k)$. For any j with $0 \le j \le k$, let ρ_j be the permutation defined by*

$$\rho_j(g_i) = \begin{cases} g_{j-i} & for\ i \le j, \\ g_i & for\ i > j, \end{cases}$$

so that $\rho_j(G) = (g_j, \ldots, g_0, g_{j+1}, \ldots, g_k)$. Then $\rho_j(G)$ is a smooth (but not necessarily compound) sequence with leading term g_j. The corresponding c values of $\rho_j(G)$ are $(b_j, \ldots, b_1, a_{j+1}, \ldots, a_k)$.

Proof For notation, let $\widehat{G} = \rho_j(G)$. Then $\widehat{G} = (\widehat{g}_0, \ldots, \widehat{g}_k) = (g_j, \ldots, g_0, g_{j+1}, \ldots, g_k)$. In what follows, we will compute $\widehat{d}_i = \gcd(\widehat{g}_0, \ldots, \widehat{g}_i)$ and $\widehat{c}_i = \widehat{d}_{i-1}/\widehat{d}_i$.

We find that

$$\widehat{d}_i = \gcd(\widehat{g}_0, \ldots, \widehat{g}_i) = \begin{cases} b_1 \cdots b_{j-i} a_{j+1} \cdots a_k & if\ 0 \le i \le j, \\ a_{i+1} \cdots a_k & if\ j < i \le k, \end{cases}$$

and

$$\widehat{c}_i = \widehat{d}_{i-1}/\widehat{d}_i = \begin{cases} b_{j-i+1} & if\ i \le j, \\ a_i & if\ i > j. \end{cases}$$

In other words, the c values for $\widehat{G}$ are $(\widehat{c}_1, \ldots, \widehat{c}_k) = (b_j, \ldots, b_1, a_{j+1}, \ldots, a_k)$.

We now verify the smoothness condition. Let $\widehat{S}_i = \langle \widehat{g}_0, \ldots, \widehat{g}_i\rangle$ and note that $\widehat{S}_i = S_i$ for $i \ge j$. To show $\widehat{G}$ is smooth, we need to show $\widehat{c}_i \widehat{g}_i \in \widehat{S}_{i-1}$ for $1 \le i \le k$. If $i \le j$, then $\widehat{c}_i \widehat{g}_i = b_{j-i+1} g_{j-i} = a_{j-i+1} g_{j-i+1} \in \langle g_j, \ldots, g_{j-i+1}\rangle = \langle \widehat{g}_0, \ldots, \widehat{g}_{i-1}\rangle = \widehat{S}_{i-1}$. If $i > j$, then $\widehat{c}_i \widehat{g}_i = a_i g_i = b_i g_{i-1} \in \langle g_0, \ldots, g_{i-1}\rangle = S_{i-1} = \widehat{S}_{i-1}$. Thus, $\rho_j(G) = \widehat{G}$ is smooth. □

When $j = 0$, $\rho_j(G) = G$, so we see that compound sequences are smooth.

Corollary 3.11 *The compound sequence $G = G(A, B)$ is smooth with c values $(a_1, \ldots, a_k)$.*

Remark 3.12 If G is smooth, $\rho_j(G)$ is not necessarily smooth. For instance, let $G = (6, 10, 11)$ and $j = 2$. Then G is smooth (with $c_1 = 3, c_2 = 2$), but $\rho_2(G) = (11, 10, 6)$ (with $\widehat{c}_1 = 11, \widehat{c}_2 = 1$) is not smooth.

Similarly, if G is compound, then $\rho_j(G)$ is not necessarily compound. As an example, let $A = (2, 2)$ and $B = (3, 3)$. Then $G(A, B) = (4, 6, 9)$ is compound. Let $j = 1$. Then $\rho_1(G(A, B)) = (6, 4, 9)$, which is not compound. However, note that $\rho_1(G(A, B))$ is smooth (which it must be by Proposition 3.10).

Since $\rho_j(G)$ is smooth by Proposition 3.10, we can apply Corollary 3.4, Lemma 3.5, and Proposition 3.6 to $\rho_j(G)$ to obtain the following.

Corollary 3.13 *Let $G = (g_0, \ldots, g_k)$ be a compound sequence. Then $G = G(A, B)$ for a suitable pair of k-tuples (A, B). Let $S = \langle G \rangle$.*

For any j with $0 \le j \le k$ and any $n \in \mathbb{Z}$, there is a unique representation $n = \sum_{i=0}^{k} n_i g_i$ where $n_0, \ldots, n_k \in \mathbb{Z}$ with $0 \le n_i < b_{i+1}$ if $0 \le i < j$ and $0 \le n_i < a_i$ if $j < i \le k$. Given such a representation, $n \in S$ if and only if $n_j \ge 0$. Furthermore, $n \in \mathrm{Ap}(S; g_j)$ if and only if $n_j = 0$.

Combining Corollary 3.7 and Proposition 3.10, we get the explicit identity for compound sequences which was shown in [4].

Corollary 3.14 [4, Theorem 3.3] *For a suitable pair (A, B) of k-tuples, let $G = G(A, B) = (g_0, \ldots, g_k)$ be the corresponding compound sequence for $A = (a_1, \ldots, a_k)$ and $B = (b_1, \ldots, b_k)$. Then for any j with $0 \le i \le k$ and any f defined on $\mathbb{N}_0$,*

$$\sum_{n \in NR} (f(n + g_j) - f(n)) = \sum_{n_0=0}^{b_1-1} \cdots \sum_{n_{j-1}=0}^{b_j-1} \sum_{n_{j+1}=0}^{a_{j+1}-1} \cdots \sum_{n_k=0}^{a_k-1} f\left(\sum_{\substack{i=0 \\ i \ne j}}^{k} n_i g_i \right) - \sum_{n=0}^{g_j-1} f(n).$$

4 Assorted Sylvester sums from free numerical semigroups

We apply Corollary 3.7 to compute certain properties of *NR*, the complement of a free numerical semigroup.

4.1 Sylvester sums

Definition 4.1 A numerical semigroup S with Frobenius number $F(S)$ is *symmetric* if for all $n \in \mathbb{Z}$ the set $\{n, F(S) - n\}$ contains exactly one element of S.

Proposition 4.2 [8, Lemma 2.14 and Corollary 4.5] *If S is a numerical semigroup, then $F(S) \le 2g(S) - 1$, with equality if and only if S is symmetric.*

If S is a free numerical semigroup, then it is symmetric, and there are explicit formulas for $g(S)$ and $F(S)$.

Proposition 4.3 [2, Theorem D] and [9, pp. 15–16] *Let S be a free numerical semigroup with smooth generating sequence $G = (g_0, \ldots, g_k)$ and corresponding c values $(c_1, \ldots, c_k)$. Then S is a symmetric numerical semigroup with*

$$F(S) = -g_0 + \sum_{i=1}^{k} (c_i - 1) g_i$$

and

$$g(S) = \frac{1}{2} \left(1 - g_0 + \sum_{i=1}^{k} (c_i - 1) g_i \right).$$

Definition 4.4 For a numerical semigroup S with generating set G, let $NR(G) = \mathbb{N}_0 \backslash S$. The mth power Sylvester sum $S_m(G)$ is defined as

$$S_m(G) = \sum_{n \in NR(G)} n^m.$$

With this notation, the genus is the 0th power sum: $g(S) = S_0(G)$. If S is symmetric (as free numerical semigroups are), then $F(S) = 2S_0(G) - 1$.

We can use Corollary 3.7 with $f(n) = n^{m+1}$ to get an explicit formula for $S_m(G)$. We give the results for the first few values of m. For notation, given a sequence $G = (g_0, \ldots, g_k)$ and some $e \in \mathbb{N}_0$, let $G^e := (g_0^e, \ldots, g_k^e)$.

Proposition 4.5 *Let $G = (g_0, \ldots, g_k)$ be a smooth sequence with corresponding c values $(c_1, \ldots, c_k)$. Then*

$$\begin{aligned} S_0(G) &= \frac{1}{2}\left(1 - g_0 + \sum_{i=1}^{k}(c_i - 1)g_i\right), \\ S_1(G) &= \frac{S_0(G)^2 - S_0(G)}{2} + \frac{S_0(G^2)}{12}, \\ S_2(G) &= \frac{2S_0(G) - 1}{6}\left(S_0G)^2 - S_0(G) + \frac{S_0(G^2)}{2}\right). \end{aligned}$$

Proof We first note that $S_0(G) = g(S)$, and the formula for $g(S)$ is given in Proposition 4.3.

For $S_1(G)$, we follow the proof of [4, Proposition 3.5] which was done for the case of G a compound sequence. Let $f(n) = n^2$. Then by Corollary 3.7,

$$\sum_{n \in NR}((n + g_0)^2 - n^2) = \sum_{n_1=0}^{c_1-1} \cdots \sum_{n_k=0}^{c_k-1}\left(\sum_{i=1}^{k} n_i g_i\right)^2 - \sum_{n=0}^{g_0-1} n^2.$$

Simplifying both sides, we have

$$\begin{aligned} g_0^2 S_0(G) + 2g_0 S_1(G) = &\sum_{n_1=0}^{c_1-1} \cdots \sum_{n_k=0}^{c_k-1}\left(\sum_{i=1}^{k} n_i g_i\right)\left(\sum_{j=1}^{k} n_j g_j\right) \\ &- \frac{(g_0 - 1)g_0(2g_0 - 1)}{6}. \end{aligned} \tag{17}$$

Since there are a finite number of summations, each with a finite number of terms, we can change the order to evaluate the summations over $n_1, \ldots, n_k$ first. We will break them up into the cases where $i = j$ and $i \neq j$. If $i = j$, we get

$$\sum_{n_1=0}^{c_1-1} \cdots \sum_{n_k=0}^{c_k-1} \sum_{i=1}^{k} n_i^2 g_i^2 = g_0 \sum_{i=1}^{k} \frac{(c_i - 1)(2c_i - 1)g_i^2}{6}.$$

If $i \neq j$, we get

$$\sum_{n_1=0}^{c_1-1} \cdots \sum_{n_k=0}^{c_k-1} \sum_{i=1}^{k} n_i g_i \sum_{j=1, j\neq i}^{k} n_j g_j = g_0 \sum_{i=1}^{k} \frac{(c_i-1)g_i}{2} \sum_{j=1, j\neq i}^{k} \frac{(c_j-1)g_j}{2}$$
$$= \frac{g_0}{4}\left(\sum_{i=1}^{k}(c_i-1)g_i\right)^2 - \frac{g_0}{4}\sum_{i=1}^{k}((c_i-1)g_i)^2.$$

Note that $\sum_{i=1}^{k}(c_i-1)g_i = 2S_0(G) + g_0 - 1$. Similarly, $\sum_{i=1}^{k}(c_i^2-1)g_i^2 = 2S_0(G^2) + g_0^2 - 1$.

The right side of Eq. (17) simplifies to

$$\frac{g_0}{12}(2S_0(G^2) + g_0^2 - 1) + \frac{g_0}{4}(2S_0(G) + g_0 - 1)^2 - \frac{(g_0-1)g_0(2g_0-1)}{6}.$$

We then subtract $g_0^2 S_0(G)$ and divide through by $2g_0$ to obtain the following.

$$S_1(G) = \sum_{n\in NR} n = \frac{S_0(G)^2 - S_0(G)}{2} + \frac{S_0(G^2)}{12}.$$

The approach for $S_2(G)$, which uses $f(n) = n^3$, is identical to the $S_1(G)$ case. Similar to the proof of [4, Proposition 3.5], we choose to omit the details here because the ideas are the same and the simplifications are more complicated. The reader is encouraged to verify the result. □

As in [4, Proposition 3.5], we note that $S_2(G)$ is a function of $S_0(G)$ and $S_1(G)$. In particular,

$$S_2(G) = \frac{2S_0(G)-1}{3}\left(3S_1(G) - S_0(G)^2 + S_0(G)\right).$$

4.2 Alternating Sylvester sums

Definition 4.6 For a numerical semigroup S with generating set G, let $NR(G) = \mathbb{N}_0 \backslash S$. The alternating mth power Sylvester sum $T_m(G)$ is defined as

$$T_m(G) = \sum_{n\in NR(G)} (-1)^n n^m.$$

In [12], for the case of $G = \{a, b\}$, the authors use Corollary 2.6 with $f(n) = (-1)^n n^m$ to find recurrences for $T_m(G)$ along with explicit formulas for $m = 0, 1, 2$. Since we must have $\gcd(G) = 1$, we may assume that b is odd and that a is either even or odd.

For notation, given non-negative integers m and g, let

$$\sigma_m(g) = \sum_{n=0}^{g} n^m \text{and} \tau_m(g) = \sum_{n=0}^{g} (-1)^n n^m,$$

the mth power and alternating mth power sums of integers in the interval $[0, g]$.

Proposition 4.7 [12, Theorem 4.1] *Let $G = \{a, b\}$.*

If a is even and b is odd, then

$$T_m(G) = \frac{1}{2}\left(\tau_m(b-1) - a^m \sigma_m(b-1) - \sum_{i=0}^{m-1} \binom{m}{i} b^{m-i} T_i(G)\right)$$

with the initial value $T_0(G) = -\frac{b-1}{2}$.

If a and b are both odd, then

$$T_m(G) = \frac{1}{2}\left(\tau_m(b-1) - a^m \tau_m(b-1) - \sum_{i=0}^{m-1} \binom{m}{i} b^{m-i} T_i(G)\right)$$

with the initial value $T_0(G) = 0$. (Note that $T_m(G)$ is symmetric in a and b when they are both odd.)

Corollary 4.8 [12, p. 1331] *When $G = \{a, b\}$, if a is even and b is odd, then*

$$\begin{aligned} T_0(G) &= -\frac{b-1}{2}, \\ T_1(G) &= \frac{1}{4}(b-1)(b-ab+1), \\ T_2(G) &= \frac{1}{12}ab(b-1)(a+3b-2ab); \end{aligned}$$

and if a and b are both odd, then

$$\begin{aligned} T_0(G) &= 0, \\ T_1(G) &= -\frac{1}{4}(a-1)(b-1), \\ T_2(G) &= -\frac{1}{4}ab(a-1)(b-1). \end{aligned}$$

We can use Corollary 3.7 to generalize Corollary 4.8 for free numerical semigroups. For $G = (g_0, \ldots, g_k)$ smooth, we use $f(n) = (-1)^n n^m$ if g_0 is odd and $f(n) = (-1)^n n^{m+1}$ if g_0 is even.

Proposition 4.9 *Let $G = (g_0, \ldots, g_k)$ be a smooth sequence with corresponding c values $(c_1, \ldots, c_k)$. Additionally, define $c_0 := 1$. Let $J := \min\{j : 2 \nmid g_j\}$ and let $I_G := \{i : 2 \mid g_i\}$. Then*

$$T_0(G) = \frac{1}{2}\left(1 - \frac{c_J g_J}{g_0} \prod_{i \in I_G} c_i\right),$$
$$T_1(G) = \frac{(2S_0(G) - 1)(2T_0(G) - 1) - 1}{4},$$
$$T_2(G) = \frac{2T_0(G) - 1}{12} \times \left(6S_0(G)^2 - 6S_0(G) + 3S_0(G^2) + g_0^2 - (c_J g_J)^2 - \sum_{i \in I_G} g_i^2 (c_i^2 - 1)\right).$$

Proof For each m, we proceed with two cases depending on whether g_0 is odd or even. We present the proof for $m = 0$ here. The proofs for $m = 1$ and $m = 2$ are similar.

Case 1 Suppose g_0 is odd. We will use $f(n) = (-1)^n$ with Eq. (16). The left side simplifies to $-2\sum_{n \in NR}(-1)^n$. To simplify the right side, first note that

$$\begin{aligned}\sum_{n_1=0}^{c_1-1} \cdots \sum_{n_k=0}^{c_k-1} (-1)^{n_1 g_1 + \cdots + n_k g_k} &= \sum_{n_1=0}^{c_1-1} \cdots \sum_{n_k=0}^{c_k-1} \left[(-1)^{n_1 g_1} \cdots (-1)^{n_k g_k}\right] \\ &= \left(\sum_{n_1=0}^{c_1-1} (-1)^{n_1 g_1}\right) \cdots \left(\sum_{n_k=0}^{c_k-1} (-1)^{n_k g_k}\right) \\ &= \prod_{i=1}^{k} \left(\sum_{n_i=0}^{c_i-1} (-1)^{n_i g_i}\right).\end{aligned}$$

The right side of Eq. (16) is then

$$\sum_{n_1=0}^{c_1-1} \cdots \sum_{n_k=0}^{c_k-1} (-1)^{n_1 g_1 + \cdots + n_k g_k} - \sum_{n=0}^{g_0-1} (-1)^n = \prod_{i=1}^{k} \left(\sum_{n_i=0}^{c_i-1} (-1)^{n_i g_i}\right) - \sum_{n=0}^{g_0-1} (-1)^n.$$

Since $g_0 = c_1 \ldots c_k$ and g_0 is odd, all of the c_i terms are odd, so the right side simplifies to $-1 + \prod_{i \in I_G} c_i$. Equating the two sides of Eq. (16), we obtain

$$\sum_{n \in NR} (-1)^n = \frac{1}{2}\left(1 - \prod_{i \in I_G} c_i\right) = \frac{1}{2}\left(1 - \frac{c_J g_J}{g_0} \prod_{i \in I_G} c_i\right),$$

with the latter equality holding because $J = 0$, which means $c_J g_J / g_0 = 1$.

Case 2 Suppose g_0 is even. By the definition of J, note that $g_0, \ldots, g_{J-1}$ are even and g_J is odd. Therefore since d_{J-1} is even and d_J is odd, $c_J = d_{J-1}/d_J$ is even. We will use $f(n) = n(-1)^n$ with Eq. (16). The left side simplifies to $g_0 \sum_{n \in NR}(-1)^n$. The right side is

$$\sum_{n_1=0}^{c_1-1} \cdots \sum_{n_k=0}^{c_k-1} (n_1 g_1 + \cdots + n_k g_k)(-1)^{n_1 g_1 + \cdots + n_k g_k} - \sum_{n=0}^{g_0-1} n(-1)^n.$$

Since $\sum_{n=0}^{g_0-1} n(-1)^n = -g_0/2$ when g_0 is even, the right side simplifies to

$$\begin{aligned}
&\left(\sum_{n_1=0}^{c_1-1} n_1 g_1 (-1)^{n_1 g_1}\right)\left(\sum_{n_2=0}^{c_2-1} (-1)^{n_2 g_2}\right) \cdots \left(\sum_{n_k=0}^{c_k-1} (-1)^{n_k g_k}\right) \\
&\quad + \left(\sum_{n_1=0}^{c_1-1} (-1)^{n_1 g_1}\right)\left(\sum_{n_2=0}^{c_2-1} n_2 g_2 (-1)^{n_2 g_2}\right) \cdots \left(\sum_{n_k=0}^{c_k-1} (-1)^{n_k g_k}\right) \\
&\quad + \cdots \\
&\quad + \left(\sum_{n_1=0}^{c_1-1} (-1)^{n_1 g_1}\right)\left(\sum_{n_2=0}^{c_2-1} (-1)^{n_2 g_2}\right) \cdots \left(\sum_{n_k=0}^{c_k-1} n_k g_k (-1)^{n_k g_k}\right) \\
&\quad + \frac{g_0}{2},
\end{aligned}$$

which equals

$$\sum_{j=1}^{k} \left(\sum_{n_j=0}^{c_j-1} n_j g_j (-1)^{n_j g_j} \prod_{i=1, i \neq j}^{k} \left(\sum_{n_i=0}^{c_i-1} (-1)^{n_i g_i}\right)\right) + \frac{g_0}{2}.$$

Since c_J is even and g_J is odd, $\sum_{n_J=0}^{c_J-1}(-1)^{n_J g_J} = 0$, so the product above is 0 for all $j \neq J$. The right side simplifies to

$$\sum_{n_J=0}^{c_J-1} n_J g_J (-1)^{n_J} \prod_{i \in I_G} c_i + \frac{g_0}{2} = -\frac{g_J c_J}{2} \prod_{i \in I_G} c_i + \frac{g_0}{2}.$$

Equating the two sides of Eq. (16), we obtain

$$\sum_{n \in NR} (-1)^n = \frac{1}{2}\left(1 - \frac{c_J g_J}{g_0} \prod_{i \in I_G} c_i\right). \qquad \square$$

Remark 4.10 Via Proposition 4.5, we can represent the formula for $T_2(G)$ in a few ways:

$$\begin{aligned} T_2(G) &= (2T_0(G)-1)\Bigg(3S_1(G) - S_0(G)^2 + S_0(G) \\ &\quad + \frac{1}{12}\left(g_0^2 - (c_J g_J)^2 - \sum_{i\in I_G} g_i^2(c_i^2-1)\right)\Bigg) \\ &= (2T_0(G)-1)\left(\frac{3S_2(G)}{2S_0(G)-1} + \frac{1}{12}\left(g_0^2 - (c_J g_J)^2 - \sum_{i\in I_G} g_i^2(c_i^2-1)\right)\right). \end{aligned}$$

In particular, since $T_0(G) \le 0$, we see that *NR* always contains at least as many odd elements as even elements. The numbers of even and odd elements in *NR* are equal exactly when all elements of G are odd.

If g_0 is odd, then $g_0 = g_J c_J$ and we can simplify the above formulas.

Corollary 4.11 *Suppose $G = (g_0, \ldots, g_k)$ is a smooth sequence.*

If g_0 is odd, then

$$\begin{aligned} T_0(G) &= \frac{1}{2}\left(1 - \prod_{i\in I_G} c_i\right), \\ T_1(G) &= -\frac{1}{4}\left(1 + (2S_0(G)-1)\prod_{i\in I_G} c_i\right), \\ T_2(G) &= -\left(\frac{3S_2(G)}{2S_0(G)-1} - \frac{1}{12}\left(\sum_{i\in I_G} g_i^2(c_i^2-1)\right)\right)\prod_{i\in I_G} c_i. \end{aligned}$$

If all terms of G are odd, then

$$\begin{aligned} T_0(G) &= 0, \\ T_1(G) &= -S_0(G)/2, \\ T_2(G) &= -S_0(G)(S_0(G)-1)/2 - S_0(G^2)/4 \\ &= -3S_2(G)/(2S_0(G)-1) \\ &= -2T_1(G)^2 - T_1(G) + \frac{1}{2}T_1(G^2). \end{aligned}$$

Suppose G contains only odd terms. As $T_2(G)$ is a function of $S_0(G)$, $S_1(G)$, and $S_2(G)$, it seems reasonable to ask whether in general $T_m(G)$ is a function of $S_0(G), \ldots, S_m(G)$ or even of $S_0(G), \ldots, S_{m-1}(G)$, since $S_2(G)$ is a function of $S_0(G)$ and $S_1(G)$. Equivalently, we ask if $T_m(G)$ is a function of $S_0(G), \ldots, S_0(G^m)$ or of $T_1(G), \ldots, T_1(G^m)$.

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